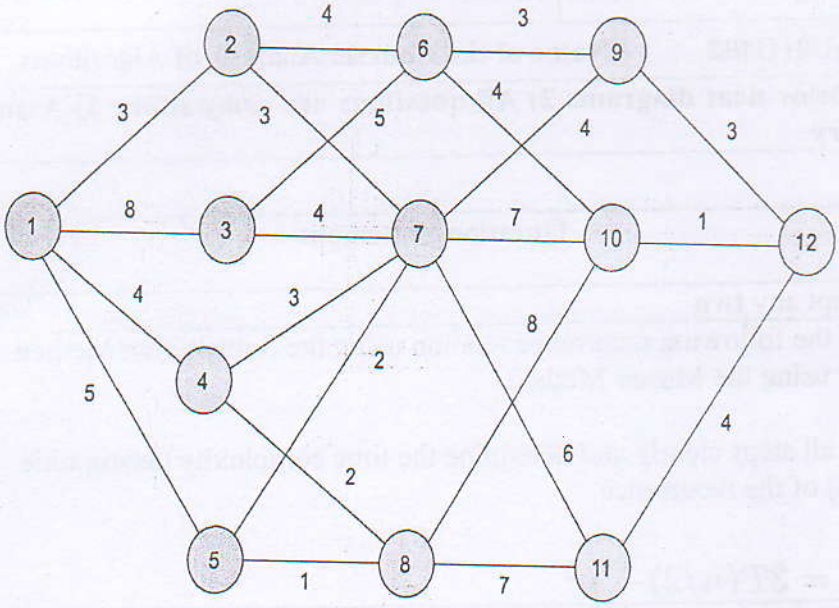
SOMAIYA VIDYAVIHAR UNIVERSITY

22·05·2025 (E)

Semester: January 2025– April 2025			
Maximum Marks: 50	**Examination: End-Semester Examination**		**Duration: 2 Hrs.**
Programme code:01 **Programme:** B.Tech in Computer Engineering	**Class:**	**SY**	**Semester:** IV (SVU 2023)
Institute/School/Department: K. J. Somaiya School of Engineering	**Name of the department:** COMP		
Course Code: 216U01C402	**Name of the Course:** Analysis of Algorithms		
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **two**	
i)	Solve the following recurrence relation using the Substitution Method and verify using the Master Method. Show all steps clearly and determine the time complexity (asymptotic bound) of the recurrence $T(n) = 3T(n/2) + n^2$	05
ii)	Explain different complexity classes with suitable examples. Comment relationship among P, NP, NP-Hard, and NP-Complete problems.	05
iii)	Solve the following string matching problem using the Longest Common Subsequence (LCS) method: Let: X = "elephant" Y = "relevant" Compute the length of the LCS and also provide the LCS string.	05
Q.2	Attempt any **one**	
i)	a) Discuss the concept of optimal substructure and overlapping sub-problems, solving the Matrix Chain Multiplication problem using the dynamic programming approach. b) Solve the following matrix chain multiplication problem using Dynamic Programming: You are given a chain of 4 matrices with dimensions array as: **P = [6, 4, 2, 4, 5]** Compute the minimum number of scalar multiplications needed to compute the product $A_1A_2A_3A_4$.	10

ii)	Solve the following multistage graphs with forward approach using dynamic programming. 1. Write the recursive formula 2. Compute the answer 3. Construct the answer 4. Write complexity  	10
Q.3	Attempt any **one**	10
i)	Solve the Traveling Salesperson problem using dynamic programming. Find a minimum cost tour starting and ending at Vertex 1. Also, write the cost of the tour. **Note:** The Adjacency matrix is given below 0 10 15 20 5 0 9 10 6 13 0 12 8 8 9 0	
ii)	During a natural disaster response operation, a team of field workers needs to transport essential supplies using a drone that has a maximum payload capacity of 20 kg. A list of items is available at the supply depot, each with its weight and a criticality score (**higher score = higher priority for survival and well-being in affected areas**). Due to drone limitations, only full units of items can be carried (no fractional items allowed).	

Available Items:

Item	Weight (kg)	Criticality Score
Food Pack	5	30
Water Containers	10	50
Medical Kit	3	60
Thermal Blanket	8	40
Flashlight Set	2	20

a) Use Dynamic Programming to compute the maximum achievable criticality score. Show the construction of the DP table and explain how the table is used to find the solution. (5 Marks)

b) Identify which items will be selected in the optimal solution and justify the selection. (4 Marks)

c) Comment on the time complexity of the dynamic programming approach used in this case. (1 Mark)

Q.4	Attempt the following	
	Solve the given problem of sum of subsets using backtracking. Give explicit, implicit and backtracking conditions for the same. N=5, M=31 Values= {11,24,7,13,31} a) Construct the **Backtracking tree and Solution tree** solve the problem using **backtracking**, showing all the feasible and infeasible nodes explored. b) Report all valid subsets (if any) whose sum equals the target sum **M = 31**.	**10**
Q.5	Attempt the following	
	Define the 8-puzzle problem. Explain the solution using the Least Cost Branch & Bound search.	**10**

1	2	3
4		6
7	5	8

Initial State

1	2	3
4	5	6
7	8	

Goal State